Fundamentals of Database

CBIO204

Biomedical

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1. Conceptual design: list of entities & attributes:

Conceptual Design

Entities & Attributes

Journal

- ISSN
- JournalName
- Impacteactor
- Quarter
- PublicherID
- Country
- StartDate
- Volume/UblicationRate
- OpenAccessStatus

Research

- ResearchiD
- Trtle
- Apstract
- PublicationDate
- Citations
- BlomedicalPldpID
- JournalID

ResearchAuthor

- ResearchID
- AuthoriD

Publisher

- PubisheriD
- PublishelName

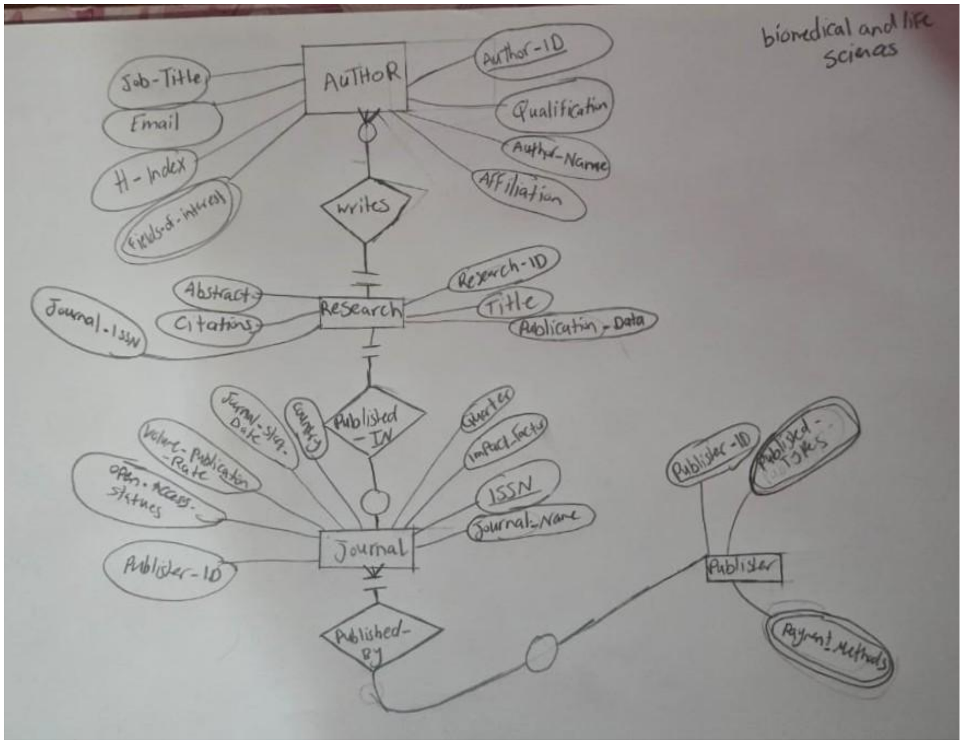
Blomedisalfteld

- FieldID

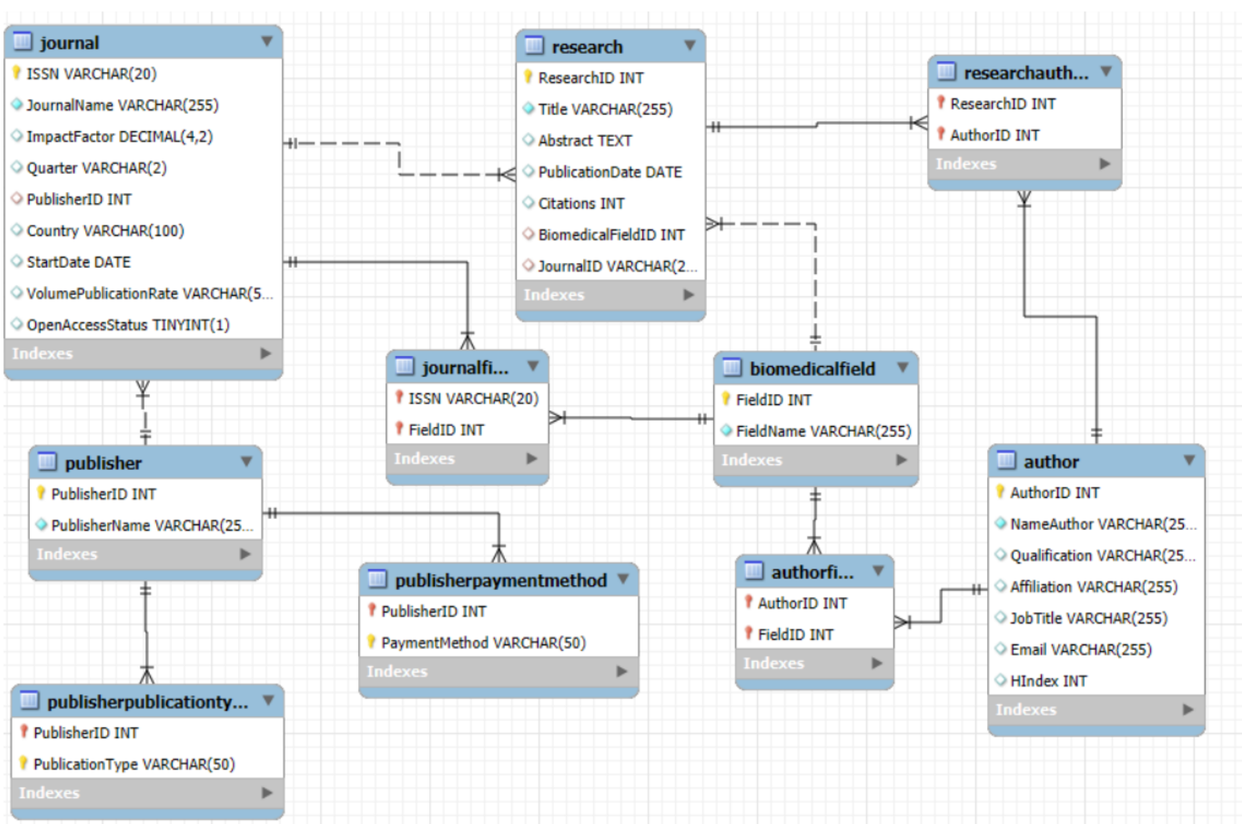
Author

- Authorib
- NameAuthor
- Qualrfication
- Aftiliation
- JobTitle

2. ER diagram: show cardinality ratios & participation constraints:



3. Relational schema: tables, attributes & constraints (PKs, FKs):



4. DDL & DML:

Database Creation

- sql code:

```
CREATE DATABASE Biomedical_1;
```

- Meaning:

Create a new database called Biomedical_1

Switch to using Biomedical_1:

- sql code:

```
USE Biomedical_1;
```

- Meaning:

Switch and use this database for the time being

Table Creation and Execution Flow:

-BiomedicalField table:

- sql code:

```
CREATE TABLE BiomedicalField (
   FieldID INT PRIMARY KEY,
   FieldName VARCHAR(255) NOT NULL
);
```

- Meaning:

Creates a table to store biomedical research fields where ‘FieldID’ is the Primary Key and ‘FieldName’ is text, required which cannot be left alone hence the usage of ‘NOT NULL’.

-Author table:

- sql code:

```sql
CREATE TABLE Author (
    AuthorID INT PRIMARY KEY,
    NameAuthor VARCHAR(255) NOT NULL,
    Qualification VARCHAR(255),
    Affiliation VARCHAR(255),
    JobTitle VARCHAR(255),
    Email VARCHAR(255) UNIQUE,
    HIndex INT
);
```

- Meaning:

It stores author details. ‘AuthorID’ is the primary key for each author. ‘Email’ must be unique (no duplicates). ‘HIndex’ is a numeric measure of research impact.

-Publisher table:

- sql code:

```sql
CREATE TABLE Publisher (
    PublisherID INT PRIMARY KEY,
    PublisherName VARCHAR(255) NOT NULL
);
```

- Meaning:

It holds publisher information. ‘PublisherID’ is primary key for each publisher.

-Journal table:

- sql code:

```sql
CREATE TABLE Journal (
```

```
    ISSN VARCHAR(20) PRIMARY KEY,
    JournalName VARCHAR(255) NOT NULL,
    ImpactFactor DECIMAL(4,2),
    Quarter VARCHAR(2),
    PublisherID INT,
    Country VARCHAR(100),
    StartDate DATE,
    VolumePublicationRate VARCHAR(50),
    OpenAccessStatus BOOLEAN,
    FOREIGN KEY (PublisherID) REFERENCES Publisher (PublisherID)
);
```

- Meaning:

It stores journal details. 'ISSN' is the primary key. 'PublisherID' links each journal to its publisher which is 'FOREIGN KEY' and ensures referential integrity. 'ImpactFactor' is a decimal value. 'OpenAccessStatus' is a Boolean (Yes/No).

-Research table:

- sql code:

```
CREATE TABLE Research (
    ResearchID INT PRIMARY KEY,
    Title VARCHAR(255) NOT NULL,
    Abstract TEXT,
    PublicationDate DATE,
    Citations INT,
    BiomedicalFieldID INT,
    JournalID VARCHAR(20),
```

```sql
    FOREIGN KEY (BiomedicalFieldID) REFERENCES BiomedicalField(FieldID),
    FOREIGN KEY (JournalID) REFERENCES Journal(ISSN)
);
```

- Meaning:

It stores research papers. It's linked to 'BiomedicalField' and 'Journal' via foreign keys. 'Citations' counts how many times the paper has been cited.

Junction Tables (Many-to-Many Relationships):

-ResearchAuthor table:

- sql code:

```sql
CREATE TABLE ResearchAuthor (
    ResearchID INT,
    AuthorID INT,
    PRIMARY KEY (ResearchID, AuthorID),
    FOREIGN KEY (ResearchID) REFERENCES Research(ResearchID),
    FOREIGN KEY (AuthorID) REFERENCES Author(AuthorID)
);
```

- Meaning:

It connects research papers to multiple authors. Composite Primary Key ensures no duplicate author-paper pair.

-AuthorField table:

- sql code:

```sql
CREATE TABLE AuthorField (
    AuthorID INT,
```

```
   FieldID INT,
   PRIMARY KEY (AuthorID, FieldID),
   FOREIGN KEY (AuthorID) REFERENCES Author(AuthorID),
   FOREIGN KEY (FieldID) REFERENCES BiomedicalField(FieldID)
);
```

- Meaning:

It links authors to multiple biomedical fields of interest.

-JournalField table:

- sql code:

```
CREATE TABLE JournalField (
   ISSN VARCHAR(20),
   FieldID INT,
   PRIMARY KEY (ISSN, FieldID),
   FOREIGN KEY (ISSN) REFERENCES Journal(ISSN),
   FOREIGN KEY (FieldID) REFERENCES BiomedicalField(FieldID)
);
```

- Meaning:

It associates journals with multiple biomedical fields.

Publisher Extensions:

-PublisherPublicationType table:

- sql code:

```
CREATE TABLE PublisherPublicationType (
   PublisherID INT,
```

```
    PublicationType VARCHAR(50), -- Journal, Book, eBook
    PRIMARY KEY (PublisherID, PublicationType),
    FOREIGN KEY (PublisherID) REFERENCES Publisher(PublisherID)
);
```

- Meaning:

It defines what types of publications a publisher produces.

-PublisherPaymentMethod table:

- sql code:

```
CREATE TABLE PublisherPaymentMethod (
    PublisherID INT,
    PaymentMethod VARCHAR(50),
    PRIMARY KEY (PublisherID, PaymentMethod),
    FOREIGN KEY (PublisherID) REFERENCES Publisher(PublisherID)
);
```

- Meaning:

It stores accepted payment methods for each publisher.

Insert:

-BiomedicalField table:

- sql code:

```
INSERT INTO BiomedicalField (FieldID, FieldName) VALUES
(1, 'Cell Biology'),
(2, 'Genetics'),
...
```

(10, 'Genomics');

- Meaning:

It adds 10 biomedical fields. Each row has a unique ‘FieldID’ and descriptive ‘FieldName’. SQL executes this by inserting one row per ‘(FieldID, FieldName)’ pair.

-Author table:

- sql code:

INSERT INTO Author (AuthorID, Name, Qualification, Affiliation, JobTitle, Email, HIndex) VALUES

(1, 'Alice Johnson', 'PhD', 'Harvard University', 'Professor', 'alice@harvard.edu', 45),

...

(10, 'Jack Wilson', 'PhD', 'University of Chicago', 'Postdoc', 'jack@uchicago.edu', 12);

- Meaning:

It adds 10 authors with details like name, qualification, affiliation, and email. ‘Email’ must be unique and SQL enforces this constraint. ‘HIndex’ is a numeric measure of research impact.

-Publisher table:

- sql code:

INSERT INTO Publisher (PublisherID, PublisherName) VALUES

(1, 'Springer'),

...

(10, 'Royal Society of Chemistry');

- Meaning:

It adds 10 publishers. Each publisher has a unique ‘PublisherID’.

-Journal table:

- sql code:

```sql
INSERT INTO Journal (ISSN, JournalName, ImpactFactor, Quarter, PublisherID, Country, StartDate, VolumePublicationRate, OpenAccessStatus) VALUES

('J001', 'Nature Genetics', 45.67, 'Q1', 5, 'UK', '1990-01-01', 'Monthly', 1),

...

('J010', 'Pharmacology Review', 9.45, 'Q4', 4, 'USA', '1980-01-01', 'Quarterly', 0);
```

- Meaning:

It adds 10 journals. Each journal is linked to a publisher via ‘PublisherID’. ‘ImpactFactor’ is a decimal value. ‘OpenAccessStatus’ is Boolean (1 = Yes, 0 = No).

-Research table:

- sql code:

```sql
INSERT INTO Research (ResearchID, Title, Abstract, PublicationDate, Citations, BiomedicalFieldID, JournalID) VALUES

(1, 'Gene Editing Advances', 'Study on CRISPR applications.', '2020-05-01', 120, 2, 'J001'),

...

(10, 'Genomic Sequencing', 'Advances in sequencing tech.', '2023-06-18', 200, 10, 'J007');
```

- Meaning:

It adds 10 research papers. Each paper links to a **BiomedicalField** and a **Journal** via foreign keys. SQL checks that ‘BiomedicalFieldID’ and ‘JournalID’ exist before inserting.

Junction Tables (Many-to-Many Relationships):

-ResearchAuthor table:

- sql code:

```sql
INSERT INTO ResearchAuthor (ResearchID, AuthorID) VALUES

(1,1),(1,2),(2,3),(2,4),(3,5),(3,6),(4,7),(4,8),(5,9),(5,10);
```

- Meaning:

It connects research papers to authors. Example: Research 1 is authored by Author 1 and Author 2. SQL enforces that both IDs exist in their respective tables.

-AuthorField table:

- sql code:

INSERT INTO AuthorField (AuthorID, FieldID) VALUES

(1,2),(2,3),(3,4),(4,5),(5,1),(6,6),(7,7),(8,8),(9,9),(10,10);

- Meaning:

It links authors to their biomedical fields of interest. Example: Author 1 is interested in Genetics (Field 2).

-JournalField table:

- sql code:

INSERT INTO JournalField (ISSN, FieldID) VALUES

('J001',2),('J002',1),('J003',3),('J004',4),('J005',9),

('J006',8),('J007',10),('J008',7),('J009',6),('J010',5);

- Meaning:

It associates journals with biomedical fields. Example: Journal J001 covers Genetics (Field 2).

Publisher Extensions:

-PublisherPublicationType table:

- sql code:

INSERT INTO PublisherPublicationType (PublisherID, PublicationType) VALUES

(1,'Journal'),(2,'Journal'),(3,'Book'),(4,'Journal'),(5,'Journal'), (6,'Book'),(7,'Book'),(8,'Journal'),(9,'Journal'),(10,'Journal');

- Meaning:

It defines what types of publications each publisher produces. Example: Publisher 3 produces Books.

-PublisherPaymentMethod table:

- sql code:

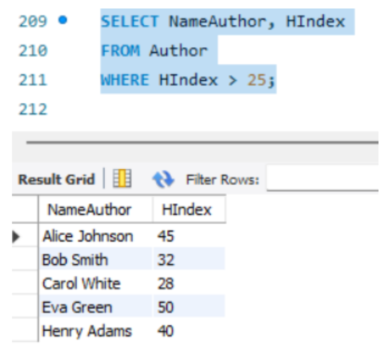
INSERT INTO PublisherPaymentMethod (PublisherID, PaymentMethod) VALUES

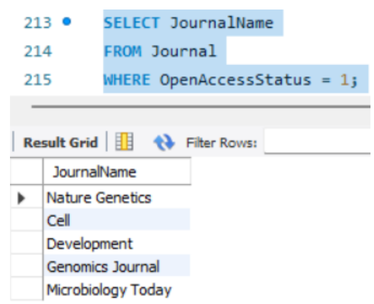
(1,'Cash'),(2,'Bank Transfer'),(3,'Online Payment'),(4,'Cash'),(5,'Online Payment'),

(6,'Bank Transfer'),(7,'Cash'),(8,'Online Payment'),(9,'Bank Transfer'),(10,'Cash');

- Meaning:

It stores accepted payment methods for each publisher. Example: Publisher 2 accepts Bank Transfer.





DML queries:

1) Select authors with h-index greater than 25:

- sql code:

SELECT NameAuthor, HIndex

FROM Author

WHERE HIndex > 25;

2) Select journals that are open access:

- sql code:

SELECT JournalName

FROM Journal

WHERE OpenAccessStatus = 1;

3) Update the impact factor of a journal:

2	Bob Smith	MD	University of California, Berkeley	Researcher	bob@stanford.edu	32

- sql code:

UPDATE Journal

SET ImpactFactor = 50.25

WHERE ISSN = 'J001';

4) Update an author's affiliation:

- sql code:

UPDATE Author

SET Affiliation = 'University of California, Berkeley'

WHERE AuthorID = 2;

5) Count total authors:

- sql code:

SELECT COUNT(*) AS TotalAuthors FROM Author;

```
SELECT J.JournalName, P.PublisherName
FROM Journal J
JOIN Publisher P ON J.PublisherID = P.PublisherID;
```

JournalName	PublisherName
Nature Genetics	Nature Publishing Group
Cell	Springer
Immunity	Elsevier
Neuron	Elsevier
Development	Cambridge University Press
Molecular Cell	Elsevier
Genomics Journal	Wiley
Biochemistry Reports	American Chemical Society
Microbiology Today	Royal Society of Chemistry
Pharmacology Review	Taylor & Francis

```
SELECT COUNT(*) AS TotalAuthors FROM Author;
```

TotalAuthors
10

6) Average citations per research paper:

- sql code:

SELECT AVG(Citations) AS AvgCitations FROM Research;

7) Show research titles with their authors:

- sql code:

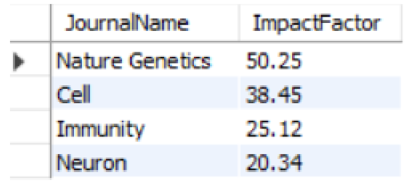
ISSN	JournalName	ImpactFactor	Quarter	PublisherID	Country	StartDate	VolumePublicationRate	OpenAccessStatus
J001	Nature Genetics	50.25	Q1	5	UK	1990-01-01	Monthly	1

SELECT R.Title, A.NameAuthor

FROM Research R

JOIN ResearchAuthor RA ON R.ResearchID = RA.ResearchID

JOIN Author A ON RA.AuthorID = A.AuthorID;



	JournalName	ImpactFactor
▶	Nature Genetics	50.25
	Cell	38.45
	Immunity	25.12
	Neuron	20.34

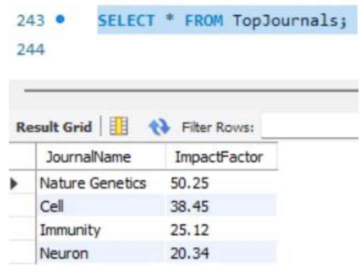
8) Show journals with their publishers:

• sql code:

SELECT J.JournalName, P.PublisherName

FROM Journal J

JOIN Publisher P ON J.PublisherID = P.PublisherID;



```
SELECT * FROM TopJournals;
```


	JournalName	ImpactFactor
▶	Nature Genetics	50.25
	Cell	38.45
	Immunity	25.12
	Neuron	20.34

9) Create a view for top journals:

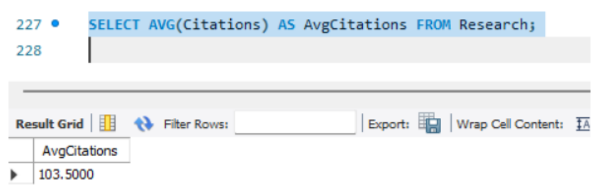
• sql code:

CREATE VIEW TopJournals AS

SELECT JournalName, ImpactFactor

FROM Journal

WHERE ImpactFactor > 20;



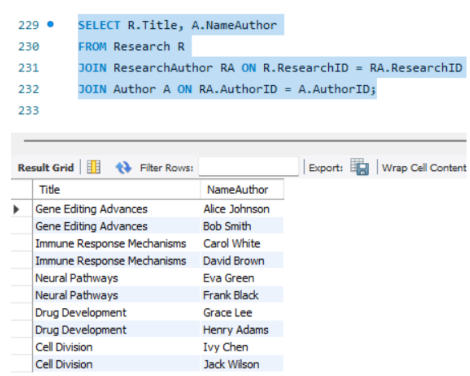
```
SELECT AVG(Citations) AS AvgCitations FROM Research;
```


	AvgCitations
▶	103.5000

10) Query the view:

• sql code:

SELECT * FROM TopJournals;



```
SELECT R.Title, A.NameAuthor
FROM Research R
JOIN ResearchAuthor RA ON R.ResearchID = RA.ResearchID
JOIN Author A ON RA.AuthorID = A.AuthorID;
```


	Title	NameAuthor
▶	Gene Editing Advances	Alice Johnson
	Gene Editing Advances	Bob Smith
	Immune Response Mechanisms	Carol White
	Immune Response Mechanisms	David Brown
	Neural Pathways	Eva Green
	Neural Pathways	Frank Black
	Drug Development	Grace Lee
	Drug Development	Henry Adams
	Cell Division	Ivy Chen
	Cell Division	Jack Wilson

ResearchID	Title	Abstract	PublicationDate	Citations	BiomedicalFieldID	JournalID
1	Gene Editing Advances	Study on CRISPR applications.	2020-05-01	120	2	J001
2	Immune Response Mechanisms	Exploring T-cell activation.	2019-03-15	85	3	J003
3	Neural Pathways	Mapping brain circuits.	2021-07-20	60	4	J004
4	Drug Development	New pharmacological targets.	2018-11-10	95	5	J010
5	Cell Division	Mechanisms of mitosis.	2017-01-05	150	1	J002
6	Microbial Resistance	Antibiotic resistance study.	2022-02-12	70	6	J009
7	Protein Folding	Biochemical analysis.	2016-09-30	110	7	J008
8	Molecular Pathways	Signal transduction research.	2020-12-01	80	8	J006
9	Developmental Stages	Embryonic growth study.	2015-04-25	65	9	J005
10	Genomic Sequencing	Advances in sequencing tech.	2023-06-18	200	10	J007
NULL	NULL	NULL	NULL	NULL	NULL	NULL

PublisherID	PublicationType
1	Journal
2	Journal
3	Book
4	Journal
5	Journal
6	Book
7	Book
8	Journal
9	Journal
10	Journal
NULL	NULL

ResearchID	AuthorID
1	1
1	2
2	3
2	4
3	5
3	6
4	7
4	8
5	9
5	10
NULL	NULL

PublisherID	PublisherName
1	Springer
2	Elsevier
3	Wiley
4	Taylor & Francis
5	Nature Publishing Group
6	Oxford University Press
7	Cambridge University Press
8	American Chemical Society
9	IEEE
10	Royal Society of Chemistry
NULL	NULL

PublisherID	PaymentMethod
1	Cash
2	Bank Transfer
3	Online Payment
4	Cash
5	Online Payment
6	Bank Transfer
7	Cash
8	Online Payment
9	Bank Transfer
10	Cash
NULL	NULL

5.Test data: at least 10 rows per table:

6. Explanation: rationale for design choices & implementation notes:

ISSN	JournalName	ImpactFactor	Quarter	PublisherID	Country	StartDate	VolumePublicationRate	OpenAccessStatus
J001	Nature Genetics	45.67	Q1	5	UK	1990-01-01	Monthly	1
J002	Cell	38.45	Q1	1	USA	1974-01-01	Monthly	1
J003	Immunity	25.12	Q1	2	USA	1994-01-01	Monthly	0
J004	Neuron	20.34	Q2	2	USA	1988-01-01	Monthly	0
J005	Development	15.89	Q2	7	UK	1953-01-01	Monthly	1
J006	Molecular Cell	18.45	Q2	2	USA	1997-01-01	Monthly	0
J007	Genomics Journal	12.34	Q3	3	USA	1985-01-01	Quarterly	1
J008	Biochemistry Reports	10.56	Q3	8	USA	1960-01-01	Quarterly	0
J009	Microbiology Today	8.90	Q4	10	UK	1970-01-01	Annually	1
J010	Pharmacology Review	9.45	Q4	4	USA	1980-01-01	Quarterly	0
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ISSN	FieldID
J002	1
J001	2
J003	3
J004	4
J010	5
J009	6
J008	7
J006	8
J005	9
J007	10

AuthorID	NameAuthor	Qualification	Affiliation	JobTitle	Email	HIndex
1	Alice Johnson	PhD	Harvard University	Professor	alice@harvard.edu	45
2	Bob Smith	MD	Stanford University	Researcher	bob@stanford.edu	32
3	Carol White	PhD	Oxford University	Lecturer	carol@oxford.edu	28
4	David Brown	PhD	MIT	Postdoc	david@mit.edu	15
5	Eva Green	PhD	Cambridge University	Professor	eva@cambridge.edu	50
6	Frank Black	MD	Yale University	Researcher	frank@yale.edu	20
7	Grace Lee	PhD	UCLA	Lecturer	grace@ucla.edu	18
8	Henry Adams	PhD	Johns Hopkins	Professor	henry@jh.edu	40
9	Ivy Chen	PhD	University of Toronto	Researcher	ivy@utoronto.edu	22
10	Jack Wilson	PhD	University of Chicago	Postdoc	jack@uchicago.edu	12
NULL	NULL	NULL	NULL	NULL	NULL	NULL

AuthorID	FieldID
5	1
1	2
2	3
3	4
4	5
6	6
7	7
8	8
9	9
10	10
NULL	NULL

FieldID	FieldName
1	Cell Biology
2	Genetics
3	Immunology
4	Neuroscience
5	Pharmacology
6	Microbiology
7	Biochemistry
8	Molecular Biology
9	Developmental Biology
10	Genomics

- Each table represents a single entity (Research, Author, Journal, Publisher, BiomedicalField).
- Redundant data is avoided by using **junction tables** for many-to-many relationships (e.g., ResearchAuthor, AuthorField).
- This ensures consistency and scalability.

-Primary Keys:

- Every table has a Primary Key (e.g., ResearchID, AuthorID, ISSN, PublisherID).
- Guarantees uniqueness of records and efficient indexing.

-Foreign Keys:

- Used to enforce relationships:
- Research → Journal (JournalID).
- Research → BiomedicalField (BiomedicalFieldID).
- Journal → Publisher (PublisherID).
- Ensures referential integrity: you cannot insert a research paper linked to a non-existent journal.

-Composite Keys:

- Junction tables (ResearchAuthor, AuthorField, JournalField, PublisherPublicationType, PublisherPaymentMethod) use **composite primary keys**.
- Prevents duplicate associations (e.g., the same author being linked twice to the same research).

-Data Types:

- ‘VARCHAR’ for text fields (names, emails, journal names).
- ‘INT’ for numeric identifiers and counts.
- ‘DECIMAL’ for impact factors (precise scientific values).
- ‘DATE’ for publication dates.
- ‘BOOLEAN’ for open access status.

-Flexibility:

- Biomedical fields are stored in a separate table, allowing easy expansion.
- Publishers can have multiple publication types and payment methods via junction tables.
- Authors and journals can belong to multiple fields, reflecting real-world complexity.

-Database Creation:

- Creates a dedicated database for biomedical research.
- Ensures all tables and queries are scoped to this database.

- Table Creation:

- Each `CREATE TABLE` defines structure, constraints, and relationships.
- SQL executes these sequentially, building the schema layer by layer.

- Foreign keys require referenced tables to exist first (e.g., Publisher must exist before Journal).

- Data Insertion:

- `INSERT INTO` populates tables with sample records.
- SQL validates constraints:
- Primary keys must be unique.
- Foreign keys must reference valid IDs.
- Unique constraints (like Author.Email) prevent duplicates.

- Data Manipulation:

- `UPDATE` modifies existing records.
- `DELETE` removes records (with caution — cascading deletes may be needed in production).
- `SELECT` retrieves data, with filters (`WHERE`), aggregates (`AVG`, `COUNT`), and joins.

- Views:

- `CREATE VIEW` stores reusable queries as virtual tables.
- Useful for analytics (e.g., HighlyCitedResearch, TopJournals).

- Joins:

- Joins connect related tables:
- Research ↔ Author (via ResearchAuthor).
- Research ↔ Journal.
- Journal ↔ Publisher.
- SQL executes joins by matching foreign keys to primary keys.